Supporting Information

The Selling closure for lattice comparison: Euclidean search keys and exact reindexing operators

Petrus H. Zwart
Molecular Biophysics and Integrated Bioimaging Division,
Lawrence Berkeley National Laboratory, Berkeley, CA 94720, USA

Computational checks for every checkable statement

Every statement in the main paper that can be verified by computation or simulation carries a mark [Sn] pointing to an entry below. Each entry states what is claimed, how to check it, and what result counts as confirmation. Function names refer to the accompanying implementation (`agentsg.cell`); the pseudo-code is enough to re-implement each check independently. Checks are of three kinds: *exhaustive* (finite enumerations that settle the claim), *symbolic* (polynomial identities), and *empirical* (measurements on the data of the paper or on random samples, which confirm reported numbers but do not prove generic statements). Section, table, lemma and appendix numbers refer to the main paper.

S1 The Selling move and termination (symbolic + random). Claim (Appendix A): the move $u_i = -v_i$, $u_j = v_j$, $u_k = v_i + v_k$, $u_l = v_i + v_l$ preserves the zero sum, permutes six vonorms, decreases u_{ij}^2 by 4ε with $\varepsilon = v_i \cdot v_j$, updates conorms by $q_{ij} = \varepsilon$, $q_{jk} = p_{jk} - \varepsilon$, $q_{jl} = p_{jl} - \varepsilon$, $q_{ik} = p_{il} - \varepsilon$, $q_{il} = p_{ik} - \varepsilon$, $q_{kl} = p_{kl} + \varepsilon$, so $\sum p$ drops by ε; and reduction terminates.

```
symbolic: with sympy, declare v0..v3 as symbolic 3-vectors, impose v0=-(v1+v2+v3),
          form u as above, expand each q_ij - stated_expression -> must be 0.
random:   for 3000 random bases B: superbase V=[-(sum B), B0, B1, B2]
             while min conorm < 0: apply move at most negative pair;
                 assert the update formulas hold to 1e-9; assert sum(p) decreased
             assert all conorms >= 0 and |det| unchanged
          report max step count (observed 103).
```

S2 Closure sizes and class counts by Voronoi type (exhaustive). Claim (Section 2.1, Table 1, Appendix A): 2/4/6/12/32 obtuse superbases in 1/2/3/3/4 classes for V1–V5; `selling_superbase_closure` reproduces the exhaustive set; the hexagonal lattice has one class.

```
for each type: pick conorms p (e.g. V1:(1,2,3,4,5,6); set the type's zeros),
    G = Gram from p, B = cholesky(G)
    brute = { {c0,c1,c2,c3} : ci in [-R..R]^3, R=2 and 3 (must agree),
              det[c1,c2,c3]=+-1, c0=-(c1+c2+c3), all pairwise dots <= 0 }
    classes = brute / (S4 relabelling of the 4x4 Gram)
    assert |brute|, |classes| == table values
    assert set(selling_superbase_closure(cell)) mod +-I == brute mod +-I
hexagonal (41.8,41.8,233,90,90,120): assert closure_class_count == 1, |closure| == 12.
```

S3 Voronoi type from the zero pattern (exhaustive). Claim (Appendix A): 0 zeros → V1; 1 → V2; 2 on opposite edges → V3; 2 sharing a vertex → V4; 3 → V5, and other three-zero patterns cannot occur.

```
for each type's test lattice: assert voronoi_type(cell) matches S2's class counts.
impossibility: for the 3-zero patterns not of cuboid form (e.g. p01=p02=p03=0),
    show algebraically v_i . v_i = 0 follows (v1 perp v0,v2,v3 with v0=-(v1+v2+v3)),
    or attempt to build a Gram with that pattern and confirm it is singular.
```

S4 Which lattice types are common (empirical, PDB). Claim (Section 2.1): monoclinic P is V4, orthorhombic P is V5, and V2–V5 cells are a large fraction of the PDB.

```
construct monoclinic P (a,b,c,90,beta,90) and orthorhombic P cells: assert types 4 and 5.
for all PDB cells: reduce, count zero conorms with a tolerance at the symmetry-
    constrained angles (exact 90/120 by convention); tabulate type by crystal system;
    report fraction with type != 1.
```

S5 One sorted key per lattice (exhaustive per lattice). Claim (Section 2.2, Appendix B): every obtuse superbase of a lattice has the same sorted key, for all five types.

```
for each test lattice of S2: keys = { sort(sqrt(conorms(C))) : C in brute closure }
    assert len(keys) == 1
also: cuboid given in odd basis (3,4,5,90,90,90) and even basis {v1,v3,v2-v1}:
    assert |sorted_root_key(A) - sorted_root_key(B)| < 1e-6   (sqrt float noise ~ 6e-8)
```

S6 Representative jumps, key does not (empirical, synthetic). Claim (Sections 1, 2.2, 3.1; Appendix B): along the monoclinic series $a = 120$, $b = 189.1$, $\beta = 91.2°$, $c : 150 \to 100$, the closure stays at 12 superbases. In the equal-perturbation test about $a = b$, reference $(40, 40, 60, 90, 91, 90)$ against ± 0.001 Å in b, the raw reduced distances are 0.080 and 118.5 and the sorted-key distances to the reference are both 0.0010. The earlier raw figure 83.7 is not the current result.

```
cells = [(120,189.1,c,90,91.2,90) for c in linspace(150,100,N)]
for consecutive pairs: dS6 = |S6(cell_k) - S6(cell_k+1)|  (raw reduced scalars, fixed order)
                       dkey = |key(cell_k) - key(cell_k+1)|
locate the boundary (max jump in dS6); report dS6 and dkey just before and after it;
assert max(dkey) <= C * step size for a constant C independent of N (refine N to test).
Note: at c ~ a the Hoelder rate applies; report dkey/step there separately.
```

S7 Lemma 1 (proof + random check). Claim (Section 2.2, Appendix B): $\|\operatorname{sort} x - \operatorname{sort} y\| = \min_{S_6} \|x - \sigma y\| \le \min_G \|x - \sigma y\|$ for G the image of S_4.

```
proof: rearrangement inequality (Appendix B).
random: for 10^4 pairs x,y in R^6_{>=0}:
    m6 = min over all 720 permutations of |x - sigma y|;  m4 = min over the 24 S4-induced ones
    assert |sort x - sort y| == m6 (1e-12) and m6 <= m4
```

S8 The key metric is Euclidean; the S_4-orbit metric is not (random, Schoenberg). Claim (Section 2.2, Appendix B): the key distance is the restriction of ℓ_2 to a convex cone; the physically exact orbit metric cannot be isometrically embedded in any Euclidean space.

```
Schoenberg criterion: a finite metric (d_ij) embeds isometrically in some R^m iff
    K = -1/2 J D2 J is positive semidefinite, D2_ij = d_ij^2, J = I - 11^T/n.
G4 = the 24 permutations of the six slots induced by relabelling the tetrahedron
     (apply each S4 element to the 2x3 coform and read off the slot permutation);
     assert no element of G4 is a single transposition (no reflections).
X = 150 random vectors in R^6_{>0}
D2_key(i,j) = |sort x_i - sort x_j|^2         -> eigmin(K) ~ 0 (>= -1e-12), rank 6
D2_orb(i,j) = min_{g in G4} |x_i - g x_j|^2    -> eigmin(K) clearly negative
observed: -2e-14 and -3.7 (against eigmax 80) respectively.
```

S9 Collision multiplicities 30/15/1/4/1 and the explicit colliding pair (exhaustive). Claim (Section 2.2, Table 1, Appendix B).

```
for 300 random 6-tuples of distinct values (V1):
    arrangements = all 720 placements into the 2x3 coform
    classes = arrangements / S4 action   -> assert 30
V2: 5 distinct nonzero values, S4-action restricted to the zero pattern -> 15; V4 -> 4;
V3, V5 -> 1.
explicit pair: coforms [[1,2,3],[4,5,6]] and [[1,2,3],[4,6,5]]:
    assert both Grams positive definite; assert not S4-related; assert equal sorted keys.
```

S10 Sorted D_7 separates the generic fibre (empirical, random). Claim (Appendix B).

```
for 300 random V1 coforms: for each of the 30 classes of S9 compute the 7 vonorms
    (v_i^2 = sum of the three conorms at vertex i; v_ij^2 = sum of the other four),
    key13 = sort(sqrt conorms) ++ sort(sqrt vonorms)
    assert all 30 key13 distinct.
Also reproduce Kurlin Ex. 6.5: coforms (5,3,4,1,1,4) and (6,3,3,2,1,3) share the
    vonorm multiset but not the conorm multiset.
```

S11 Angular noise is linear in conorms and $\sqrt{}$-amplified on zero slots (empirical, simulation). Claim (Section 2.3, Appendix C): $\sigma_{p_{ij}} \approx |v_i||v_j|\sigma_\theta$; a 0.01° error moves the $P6_3$ key by ≈ 1.4–$1.6\,\text{Å}$ versus $\approx 0.03\,\text{Å}$ for a generic cell.

```
cell = (41.8,41.8,233,90,90,120); r0 = sorted_root_key(cell)
for sigma in (0.01,0.05,0.1) deg, 300 draws: perturb the three angles by N(0,sigma);
    shift = |sorted_root_key(noisy) - r0|;  report median  (expect ~1.4-1.6, 3.1-3.3, 4.6 A)
    also record the conorm perturbation and check its sd ~ |v_i||v_j| sigma (radians)
repeat for triclinic (41.8,55.2,73.1,81,97,112): expect ~0.3 A at 0.1 deg.
```

S12 Noise reduction by the three stabilised keys (empirical, simulation). Claim (Section 2.3, Appendix C, Table 4).

```
as S11, with key in { sqrt(p), sqrt(p+s)-sqrt(s), sqrt(max(p-2s,0)), p/sqrt(T) },
    s = sigma_theta(rad) * T,  T = sum of conorms  (noise_floor(cell, sigma))
report the 3x4 table of median shifts; expect the ordering and magnitudes of Table 4.
```

S13 T is closure-invariant; per-pair floors are not (exhaustive per lattice). Claim (Appendix C).

```
cuboid A=(3,4,5,90,90,90), even basis B={v1,v3,v2-v1}:
    assert sum(conorms(A)) == sum(conorms(B)) (=50)
    for each closure member C of S2's lattices: assert sum(conorms(C)) constant
    per-pair floors |v_i||v_j|sigma: assert keys from A and B differ (~0.02 A for floored,
        ~0.37 A for linear with L=max|v_i|); with s=sigma T: assert difference < 1e-12.
```

S14 Full-closure matching recovers the automorphism coset (exhaustive per lattice). Claim (Sections 3.5, 3.7): oP vs itself $\to$ 8 operators (2 with class representatives); $P6_3$ cell vs itself $\to$ 24 (4 with representatives); oP odd vs even basis $\to$ 8 (0 with 24 relabellings only); every returned $\mathbf{P}$ is integer with $\det \pm 1$ and satisfies the metric identity.

```
ops = reindexing_via_canonical(cell_A, cell_B, boundary_rel=0)
assert len(ops) == expected; for P in ops: assert P integer, |det P|=1, P^T G_A P == G_B
representatives-only variant: match over selling_closure_representatives -> 2 / 4
independent check: enumerate all integer P with |entries|<=2, keep those with
    P^T G_A P == G_B exactly; the count must equal len(ops).
```

S15 End-to-end reindexing to a reference recovers an arbitrary resetting (exhaustive + random). Claim (Section 3.5): for a reference R and N = the same lattice in a random unimodular basis (with and without noise, with and without centering), the procedure of Section 2.4 returns a set of operators equal to $\mathbf{M} \cdot \mathrm{Aut}(R)$.

```
for R in {triclinic, monoclinic P, C-centred monoclinic, orthorhombic P, I-centred
          orthorhombic, tetragonal, hexagonal, cubic F} (exact cells):
    for 50 random unimodular M (entries in [-2,2], det +-1):
        G_N <- M^T G_R M (+ optional Gaussian noise: angles sigma_theta, lengths sigma_len)
        ops <- REINDEX_TO_REFERENCE(R, cell(G_N), sigma)
        aut <- {P : P integer, |det|=1, P^T G_R P == G_R}   (brute force |entries|<=2)
        assert set(ops) == { M0 * a : a in aut } up to the direction convention
        assert every P in ops is integer with |det P| = 1
    with space group given: assert |ops| / |L| equals the branch count and that the
        cached representatives are one per coset and of minimal order
    centred cases: additionally assert P_conv = C_R P C_N^{-1} has denominators
        dividing the centering index (2 or 4) and maps conventional metrics.
Le Page comparison: repeat the Table 3 check --- for the monoclinic family with
    |a-c| = 0..10 A, assert the exchange operator is in ops for every |a-c|.
```

S16 Which cosets H/L contain a twofold (exhaustive over crystallographic point groups). Claim (Appendix D): for index-2 pairs the non-trivial coset always contains an order-2 rotation; for higher index some cosets contain none.

```
build the proper rotation groups 1,2,222,4,422,3,32,6,622,23,432 as integer matrices
for each lattice group H in {222,422,622,432} and each subgroup G < H:
    cosets = H / G;  for each non-identity coset: has2 = any(order(x)==2 for x in coset)
    record the number of cosets without a twofold
observed: 0 for every index-2 pair (222/2, 422/4, 422/222, 622/6, 622/32, 432/23, 432/4*),
          1 for 422/2 ({4,4^3}), 1 for 622/3, 2 for 432/222, 5 for 432/2, 5 for 622/1, 15 for 432/1.
          (*432/4 has index 6 and still has a twofold in every coset.)
```

S17 Pseudo-symmetry as cosets between closures (exhaustive per lattice + synthetic families). Claim (Sections 2.5, 3.6; Appendix D): projecting R onto a nearby stratum and self-matching over the closure gives $H^* \supseteq H$ exactly; the quotient reproduces the expected pseudo-symmetry with residual linear in the distance.

```
monoclinic family (120, 189.1, 120+d, 90, 91.2, 90), d in (0, 0.5, 1, 2, 4, 10):
    H  <- self-match(R)               -> order 4 (2/m incl. -I) for d > 0
    pattern: the two near-equal conorms that become equal at a = c
    R* <- impose;  H* <- self-match(R*) -> order 8;  assert H subset H*, index 2
    assert the a<->c exchange is in H* \ H for every d
    assert residual(exchange) / d is constant to 5%   (linear in distance)
near-tetragonal cuboid (3, 3+e, 5, 90, 90, 90), e in (0.01, 0.1, 0.5):
    H order 8 (mmm); pattern p01 ~ p02; H* order 16 (4/mmm); assert index 2 and that
    the 4-fold is in H* \ H with residual ~ e
The published checks are these two hand-imposed strata. They do not enumerate every degeneration pattern.
in all cases: assert every P in H* is integer, |det|=1, and P maps the closure of R*
    onto itself (closure completeness at the stratum).
```

S18 Verification tolerance selects coset members under noise (empirical, simulation). Claim (Section 3.7).

```
two P6_3 frames with 0.05 deg angular noise (fixed seed):
    for verify_rel in (1e-6,1e-4,1e-3,1e-2): n = len(reindexing_via_canonical(A,B,...))
    expect 0, 12, 24, 24; the 12 is the noise-selected half coset.
```

S19 Noisy $P6_3$ frames are classified V4 and give the full coset with the angular tolerance (empirical, simulation). Claim (Section 3.7).

```
for sigma in (0.01,0.05) deg: A,B = independently perturbed frames
    assert voronoi_type(A, angle_sigma=sigma) == 4
    assert len(reindexing_via_canonical(A,B,verify_rel=1e-3,angle_sigma=sigma)) == 24
```

S20 Table 3: Le Page gate drops the exchange, closure recovers it (empirical, synthetic). Claim (Sections 2.4, 3.6; Table 3).

```
family: monoclinic (a, b, c=a+d, 90, beta, 90) for d in (0,1,2,2.85,4,5,10)
setting 2 = axes a<->c exchanged (exact integer relation P0 = [[0,0,1],[0,1,0],[1,0,0]] up to sign)
    delta = Le Page angular residual of P0 on the metric (2-fold axis deviation)
    gate  = delta <= 3 deg
    assert P0 (or a coset member) in reindexing_via_canonical(setting1, setting2) for every d
    report sorted-key distance between the two settings (expect ~1e-14).
```

S21 6960-matrix brute force vs a cached coset on 200 frames (empirical, benchmark). Claim (Sections 2.4, 3.5): the reference closure is enumerated once. A rerun gives median 15 ms per frame for the 6960-matrix baseline and median 75 ms per frame for the cached procedure. The earlier figures 0.07 ms and 330-fold are not this measurement.

```
200 perturbed monoclinic frames straddling the reduction flip (fixed seed, stored):
    brute: for each frame, test all unimodular 3x3 with entries in {-1,0,1} (6960)
           against the reference metric with the 2%/2deg gate
    cached: PREPARE_REFERENCE once; per frame REINDEX_FRAME
    assert the per-frame stage never re-enumerates the reference closure (call count = 1)
    assert the cached result is the two-operator coset
```

S22 When a $\{-1,0,1\}$ unimodular search is complete (stratum-wide machine proof + random + adversarial + implementation check). Claim (Sections 2.2, 2.3, 4.3; Appendix D): (i) most random resettings need entries of magnitude 2 and are missed on unreduced input; (ii) between Niggli cells of one lattice every relating operator has entries in $\{-1,0,1\}$ (exact and noisy, eight lattice types, plus adversarial anisotropic and near-degenerate cases); (iii) proposition, machine-checked per Voronoi type and valid for every lattice: the closure in one superbase's integer coordinates is stratum-fixed (1/2/3/6/16 classes) and obtuse on the whole open stratum; between any two Selling-reduced settings some relating operator has entries in $\{-1,0,1\}$; the full coset escapes — V4 self-cosets reach entry 2 (18/24 bases), V5 reach 2 (60/64) and cross-class pairs reach 3 — while V1–V3 cosets never do (an earlier draft placed the escape in V3; that construction's counterpart cell was not reduced, and the corrected enumeration retires it); (iv) the Niggli reducer's change-of-basis must satisfy the metric identity; (v) between different lattices the radius-one truncation fails where the certificate radius exceeds 2.

```
(i)   200 random unimodular M with entries in [-2,2]: count max|entry| > 1  (observed 192/200)
(ii)  for R in {oF(3,4,5), oI(3,4,5), cF, cI, hP(3,3,5), mP(120,189.1,120.3,90,91.2,90),
       oF(3,3.2,5), triclinic(40,50,60,80,95,110)} (centred ones made primitive):
       trials: two copies, noise (0.02 A, 0.05 deg) BEFORE random resettings M1, M2
       (entries in [-2,2]); Niggli-reduce both; coset ops between the reduced cells:
       max|entry| = 1 in all cases, also without noise.
       adversarial (exact): needle hP(41.8,41.8,233), plate(80,80,3.4,gamma=120),
       near-tie(10,10.001,10.002,89.99,90.01,90), near-cube, cuboid(3,4,5),
       skew needle(7.47,47.69,7.15,91.69,90.81,98.53); direct operator
       P = (M2 N2)^{-1}(M1 N1): integer, metric identity to 1e-6, max|entry| = 1.
(iii) per type V1..V5: Gram linear in the stratum's free conorms ->
       flip-BFS with integer conorm-COEFFICIENT vectors (a slot is zero iff its
       coefficient vector is zero) gives the stratum-fixed closure:
       1/2/3/6/16 classes; every member conorm has coefficients >= 0 (obtuse on
       the whole open stratum). Generic automorphisms: Q^T Gk Q = Gk for all
       coefficient matrices (orders 2/2/2/4/8, entries <= 1). For all member-basis
       pairs (A,B), coset = B^-1 Aut A:
         min over coset of max|entry| = 1 for every pair, every type;
         max: V1/V2/V3 = 1; V4 = 2 (self-pairs: 18 of 24 bases escape);
              V5 = 3 (self-pairs 60/64 at 2; cross-class pairs at 3).
       numeric instance (cuboid 3x4x5): settings (7.07,4,5,90,135,124.45) and
       (6.40,4,5.83,90,132.04,128.66); coset of 8, entries [1,1,1,1,2,2,3,3];
       the {-1,0,1} scan finds exactly 4; the closure route returns all 8
```

```
       (and the 4-operator V4 self-coset with its entry-2 members).
       regression: the retracted V3 'entry-2 pair' came from pairing closure
       members under arbitrary signed permutations; the constructed cell's
       superbase completion has a negative conorm, i.e. it was not reduced.
(iv)  for random cells: (red, N) = niggli_reduce(cell); assert N^T G N == G(red) to 1e-9,
       |det N| = 1, and convergence on noisy cells. On the current reducer a 500-cell
       noisy sample (seed SEED_BRUTE+1) gave 0 non-convergences and maximum metric
       error 8e-13. The earlier report that an in-house reducer failed the identity,
       and failed to converge on 1--2 percent of noisy cells, is not reproduced.
       The Selling route does not use this reducer.
(v)   reduced needle pair A=(7.47,47.69,7.15,91.69,90.81,98.53),
       B=(7.27,42.89,7.01,91.26,90.24,92.72): ball(1) minimum 298.2, ball(2)
       minimum 105.1 (ratio 2.8), minimiser entry 2, certificate radius
       R(D) = 7.0 > 2 -- the truncation fails exactly where eq. (cert) says so;
       random search: 35 of 400 anisotropic reduced pairs improved at radius 2.
```

S23 KD tree on 206 214 PDB keys returns planted neighbours (empirical, benchmark). Claim (Section 3.2): on the precomputed keys, cKDTree build is 0.044 s and the median $k = 10$ query is 0.010 ms (Python 3.12, NumPy 2.5.3, SciPy 1.18.1). Earlier figures of 0.28 s and 0.04 ms are not this measurement. A radius query returns a planted neighbour.

```
keys = [sorted_root_key(cell) for cell in PDB]   (report which key variant, exactly)
tree = cKDTree(keys)                             # 206214 six-vectors
on a 3000-cell subset, plant a neighbour inside the query radius;
    assert every planted neighbour is returned by the radius query (Lemma 1).
```

S24 Crystal systems occupy coherent regions of the shape embedding (empirical, PDB). Claim (Section 3.2) is qualitative; the check is that the embedding is reproducible and that the labels, unused in its construction, are locally homogeneous.

```
k-NN graph on keys/V^(1/3); 2-D embedding with a fixed seed.
homogeneity: for each cell, fraction of its k neighbours (in key space, not 2-D)
    with the same crystal system; report the distribution and compare with the
    label-shuffled null.  Reproducibility: repeat with 3 seeds; report Procrustes distance.
```

S25 Collision / edge audit of the key-space graph (empirical, PDB). Claim (Section 3.2): the fraction of edges whose key distance is within 10% of the exact orbit distance has not been reported. The check below is the computation that would report it. Do not read embedding branches as topology until that fraction is in the paper.

```
for each k-NN edge (i,j): ops = reindexing_via_canonical(cell_i, cell_j, verify_abs=calibrated)
    classify: same lattice (ops non-empty), collision (keys within eps but ops empty),
              ordinary neighbour (otherwise)
report counts; report the fraction of edges whose key distance is within 10% of the
    exact orbit distance computed over the closure (edge fidelity).
```

S26 Local entropy effective rank of key neighbourhoods (empirical, PDB). Claim (Section 3.2): $k = 40$ neighbourhoods on a 30 000-cell subsample of the volume-normalised keys give entropy effective rank from 1.00 to 5.69, median 2.02. The previously printed median 4.12 is not reproduced by this estimator.

```
for each of M neighbourhoods (k nearest in key space, k stated): X = centred keys (k x 6)
    sigma = singular values; p = sigma/sum(sigma); r_eff = exp(-sum p log p)
report the range over occupied neighbourhoods; state k and M.
low-rank filaments sit near one; higher-rank neighbourhoods approach six.
```

S27 Lysozyme census numbers and the shrinkage coordinate (empirical, PDB). Claim (Sections 2.6, 3.3): 1372 entries / 1361 with cells; 1152 $P4_32_12$, 100 $P2_12_12_1$, 59 $P2_1$, 41 $P1$; 1113 native-volume tetragonal cells spanning 26% in volume; leading intrinsic coordinate vs shrinkage $|r| = 0.92$; $a \in [76.8, 80.0]$; c steps near 37/38.

```
entries = PDB search UniProt P00698 (record the date); count cells and space groups.
tetragonal subset: V = a^2 c; native-volume filter as stated; (max V - min V)/median V.
graph = k-NN on keys; coordinate 1 = leading eigenvector of the graph Laplacian
    (or first diffusion coordinate); r = Pearson(coordinate, V^(1/3)).
scatter a and c against the coordinate; report ranges and the c-value histogram.
```

S28 CXIDB 83 stream statistics and the orientation diagnostic (empirical). Claim (Section 3.4): 14 445 patterns, 12 474 crystals, $P6_3$; c median 233.3, MAD 0.13, maximum 245 Å. Beam within 10° of c^* ($n = 141$): c MAD 0.22 Å. Beam angle above 80° ($n = 2866$): c MAD 0.083 Å. Ratio 2.6. The earlier scatters 1.53 Å and 0.5 Å are not this binning.

```
parse stream: count chunks and 'Cell parameters' records; read per-crystal cell and
    orientation matrix (astar,bstar,cstar).
c distribution: median, MAD, 99.9th percentile.
keys = sorted conorms per crystal (linear key); PCA on centred keys; report loadings.
theta = angle between beam direction and cstar; bins: theta<10 deg vs theta>80 deg;
    report MAD or sd of c in each bin; bootstrap CI on the ratio (expect ~3).
```

S29 The linear archive key: invariance, scaling, Lipschitz, rearrangement bound, pseudo-symmetric fidelity (exhaustive per lattice + synthetic + deposited pairs). Claim (Sections 2.2, 2.3; Appendix C): one key per lattice over the full closure; sort$(p/\sqrt{T})$ scales as length with the cube identity; Lipschitz where the square-root key is Hölder-$\frac{1}{2}$; the sorted distance lower-bounds the aligned distance over the 24 relabellings; between pseudo-symmetric lattices the linear distance scales as δ where the square-root distance scales as $\sqrt{\delta}$; on the Scotty pairs the linear key reproduces NCDist's ranking and the square-root key reverses it — NCDist values carry an arbitrary internal scale factor and are used only ordinally (cells and NCDist values from McCoy *et al.*, 2026, Tables 1–5).

```
closure invariance: for each V1-V5 test lattice, every member of the brute-force
    closure gives the same sort(p/sqrt(T))            (one key, all five types)
scaling: key(s*cell) = s*key(cell); cube of edge a: key = (0,0,0,a/sqrt(3) x3)
    and |key(a+d) - key(a)| = d exactly
Lipschitz: P63 cell (41.8,41.8,233,90,90,120), angle noise 0.01 / 0.1 deg:
    median key shift 0.011 / 0.12 A   (square-root key: 1.50 / 4.47 A)
rearrangement bound: 300 random cell pairs: sorted linear distance <= min over
    the 24 relabellings of the aligned linear distance   (0 violations)
pseudo-symmetry: monoclinic (50,70,60,90,beta,90), pairs (90+d, 90+2d) for
    d = 1, 0.1, 0.01 deg: linear 0.80 / 0.080 / 0.008 A
    against square root   3.1 / 0.95  / 0.30  A
Scotty cells: 1fe5->1u4j 2.72, 1fe5->1g2x 2.87, 4nl7->2yht 5.43 A
    (same ranking as NCDist 2.09 / 2.08 / 2.34 -- arbitrary scale, ordinal
    use only; the square-root key 16.7 / 16.4 / 13.8 A reverses the order)
```

S30 The Bravais skeleton of the normalised key space: points, nexus, seams (exhaustive per structure + coarse scan). Claim (Section 4.1, Table 5): the three cubic lattices are fixed points; cP is a quintuple nexus; hR meets tI at exactly cI and cF, fed by oI and oF respectively; hP runs beside hR under the 3:1 aspect pairing and tP beside tI under the 2:1 pairing; the tetragonal curves are farthest apart near the cubic points.

```
normalised key: k = sort(p / sqrt(T)) / V^(1/3),  T = sum of conorms
fixed points: cP = (0,0,0,0.5774 x3); cI = (0.2572 x6); cF = (0,0,0.3969 x4)
quintuple nexus at cP: tP(c/a -> 1) approaches linearly; oP corner; oC(a=b)
    coincides with tP exactly and oC(a,a,a/sqrt2) -> cP (< 1e-8); hR(alpha -> 90)
    approaches linearly
double nexus: hR meets tI only at (alpha = 109.471, c/a = 1) = cI and at
    (alpha = 60, c/a = sqrt2) = cF; a coarse scan of both curves (alpha 40..115,
    c/a 0.4..2.8) finds no other contact below 0.05
```

```
feeds: oI(a=b) = tI exactly and oI(a=b=c) = cI;  oF(a=b) = tI(sqrt2 c/a)
    exactly and oF(a=b=c) = cF  (the F222 side)
seam hP || hR (3:1): d( hP(R), hR(hex c/a = 3R) ) = 0.093, 0.014, 0.0022,
    0.0009 at R = 1, 4, 16, 32  (monotone)
seam tP || tI (2:1): d( tP(R), tI(2R) ) = 0.64, 0.15, 0.025, 0.004 at
    R = 1, 4, 16, 64; flat branch d( tP(R), tI(R/sqrt2) ) -> 0 as R -> 0
separations: d(cP, tI curve) = 0.50; d(cI, tP curve) = 0.71
```

S31 The closure minimum is a congruence statistic, not a similarity metric (random + constructed + deposited pairs). Claim (Sections 2.2, 2.4; Appendix D): the key lower-bounds the closure-aligned minimum; between different lattices that minimum is a face-discontinuous upper bound; widened matches are not bounded by the key; the interior margin condition certifies exactness.

```
random pairs (N = 120, seeded): sorted key <= closure minimum in every case
    (0 violations); equality in 1 of 120; ratio median 1.14, max 4.9
near-face pair, p_A = (4,7,9,5,6,+delta) and p_B realized from
    (4,7,9,5,6,-delta): key ~ 0.165 / 0.033 / 0.007 at delta = 0.5 / 0.1 / 0.02;
    closure minimum 0.536 / 0.510 / 0.508 (stuck); widened (rel 0.2)
    0.184 / 0.037 / 0.0074  -> closure / widened = 3 / 14 / 71
Hfq 4nl7 -> 2yht: key = closure minimum = 5.434; widened minimum 5.434,
    4.455, 2.825, 2.651 at boundary_rel = 0.02, 0.06, 0.12, 0.3
    (same direction as NCDist, whose value 2.34 is on an arbitrary internal
    scale and is not compared by magnitude; McCoy et al., 2026)
interior pairs (N = 40): widening improved the minimum in 3 cases; assert
    that in each such case the least positive normalised conorm of one
    lattice lies below the minimum (the certificate's contrapositive)
```

S32 Guarantee under the noise model: retrieval, coverage, coset, soundness (all five Voronoi types + failure onset). Claim (Sections 2.4; Appendix D): with radius, zero tolerance and residual at stated multiples of one conorm noise scale, two noisy measurements of one lattice are retrieved and yield the full exact coset; setting a tolerance below its multiple produces the partial-coset failure.

```
for each Voronoi type V1..V5 test cell, angular noise sigma = 0.02 deg,
    fixed seed, two independently noised copies:
    expected = coset size of the clean cell matched against itself
    with angle_sigma = sigma and verification rel = 1e-3, assert the
    recovered operator count equals expected
    retrieval: assert the noisy pair's linear-key distance is below the
    noise-scale radius
failure onset (P6_3 cell, sigma = 0.05 deg): operator counts at
    verification rel = 1e-6, 1e-4, 1e-3 are non-decreasing; the smallest
    tolerance recovers a strict subset (none, at this seed) and the
    noise-scale tolerance recovers the full coset of 24 [Section 4.3]
```

S33 The ideal: certified, symmetrised deformation distance (constructed + deposited pairs + random ladder check). Claim (Section 2.2, Table 6): the minimum-norm deformation to congruence is attained and certified by the data-sized integer ball; the minimiser is nearly obtuse — at value D its frame's superbase conorms in the anchor's reduced coordinates are $\geq -\sqrt{3}D$, which collapses the $\mathrm{GL}(3,\mathbb{Z})$ minimisation into the widened closure; it is continuous through Selling faces; the directed values are asymmetric and the report symmetrises them; every ladder rung upper-bounds it.

```
implementation: agentsg.cell.deformation.deformation_distance
ball counts: entries <= 1 -> 6 960 unimodular of 19 683 candidates;
    entries <= 2 -> 135 408
sufficiency of the radius-one ball: iff lmax(G_A) + D < 4 lmin(G_B)
```

```
    (R(D) < 2); Hfq pair R = 1.86 (suffices); the P6_3 cell gives
    R = 7.9 already at D = 0, where the widened closure is the
    enumerator instead
near-face pair (one conorm at +delta vs -delta, delta = 0.1 and 0.02):
    d = 4*delta exactly (Frobenius on G); both directions equal; certified
    at radius 1; Delta_cell = one angle by -8.54*delta degrees
Hfq 4nl7 -> 2yht: bound ||M|| <= 1.86 so the entries-<=1 ball suffices;
    directed values 492 and 533 A^2, both certified; reported d = 533
    (the max); the 8% difference is the anchoring asymmetry
ladder: for 15 random near pairs, the jointly-obtuse closure match
    evaluated in the same norm upper-bounds the certified ideal
near-obtuseness lemma: random reduced pairs (seeded; anchors verified
    obtuse in the standard superbase coordinates), global minimum over the
    entries<=2 ball at value D: the minimising frame's superbase conorms
    are >= -sqrt(3) D in the anchor's coordinates, from the bilinear bound
    |w_i^T E w_j| <= |w_i||w_j| ||E||, max |w_i||w_j| = sqrt(3);
    observed worst penetration 0.70 D over 40+ pairs
```

S34 Relating reduced cells under a declared error radius (Scotty bases + seeded sweeps). Claim (Sections 2.2, 4.4, Table 7): with the error model $\mathbf{G}' = \mathbf{G}_{\text{Niggli}} + D\,E$, E random symmetric with $\|E\|_F = 1$ (the ideal's norm, so D, $\sqrt{3}D$ and $R(D)$ are commensurate), the Niggli basis jumps chambers in more than half of directions already at $D = 20\,\text{Å}^2$ on 2yht; the radius-one ball and the widened closure both recover a value $\leq D$ in every run; the ball's certificate $R(D) < 2$ is the anisotropy budget $D_{\max} = 4\lambda_{\min} - \lambda_{\max}$, negative for 7 of the 17 Scotty bases (both lysozyme cells, 5wey, and all four GabD cells), on which it certifies nothing at any error level; the widening certificate at $\sqrt{3}D$ is anisotropy-independent.

```
budgets: for each Scotty base (primitive, Niggli-reduced): D_max = 4 lmin - lmax
    of the Niggli Gram; assert the negative set is exactly
    {2d4k, 1vdp, 5wey, 4nl6, 1jr7, 2r6s, 8cdf}; 2yht in (1600, 1900);
    4amt in (800, 1000); 2d4k < -900                        [Table 7]
sweep (2yht, D = 20, seeded, n = 8..25): perturb G_N by D*E; Niggli-reduce;
    jump = CoB not a signed permutation (observed 14/25);
    ball-1 minimum <= 1.05 D (25/25) and R(min) < 2 (25/25);
    widened closure at boundary_rel = sqrt(3) D / max vonorm:
    matched metric residual <= 1.05 D (24/25)
sweep (2d4k, D = 20, seeded, n = 5..15): ball-1 minimum <= 1.05 D (15/15)
    but R(min) >= 2 in every run (0/15 certified -- the budget is negative);
    widened closure residual <= 1.05 D (15/15), certificate intact
anisotropic column search (P6_3 needle (41.8, 41.8, 233, 90, 90, 120) vs
    itself, D = 50): per-column shells |m_j^T G m_j - G_jj| <= D hold
    6 / 6 / 2 vectors (72 column triples) against the entry box 15^9 at
    floor(R) = 7; the in-plane norm-form-31 stack (length 232.7 A) sits
    124.6 A^2 below c^2 = 54289 and enters the long shell only at D ~ 125,
    where the wide count exceeds 2 -- the pseudo-sublattice ambiguity
    appearing exactly where the per-column prescription admits it
```

S35 Carbonic anhydrase II: one protein, one deformed family (1065 deposited cells). Claim (Section 4.5): the 1065-entry carbonic anhydrase list is retrieved by one sorted-key ball of radius 2.5 Å around 1CA2; it is a single continuously deformed crystal form (1062 $P2_1$ + 3 $P1$, one component at radius 0.5 Å); the medoid 4L5V has positive anisotropy budget ($\approx 1155\,\text{Å}^2$); on seeded pairs the radius-one ideal is certified and the interior margin makes the closure minimum exactly the ideal; strains run from $\sim$1% (median) to 10.3% (6BBS, certified at $639\,\text{Å}^2$).

```
data: agentsg/tests/data/ca_cells.csv -- pdb, sg, cell for the user-supplied
    1065 codes; cells and space groups from the PDB snapshot
    (doi 10.5281/zenodo.22986222); harvested by one sorted-linear-key search
```

```
    around 1CA2 (P 1 21 1, 42.7 41.7 73 90 104.6 90) at cutoff 2.5 A;
    every listed code fell inside the ball (recall 1.0 from one query);
    the ball also holds 145 entries not on the list (not asserted).
family: sg counts {P 1 21 1: 1062, P 1: 3}; 956 distinct deposited cells;
    NN key distance median < 0.1 A (observed 0.05); union-find on key
    pairs at 0.5 A gives one component >= 1060 members containing the
    three P1 deposits.
medoid: argmin of summed key distances = 4L5V (42.361 41.398 72.235
    90 104.37 90); Niggli budget D_max = 4 lmin - lmax in (1000, 1300)
    (observed 1155 A^2).
pairs (seeded, n = 25; a 60-pair run gives the quoted statistics):
    radius-1 ideal certified (R < 2 both directions) 25/25;
    interior margin (closure minimum below every positive normalised
    conorm of both cells) 25/25 -> closure minimum == ideal;
    d median in (40, 130) (observed 82; p90 154; max 209 A^2);
    strain median 0.012, p90 0.022.
extreme: deformation_distance(4L5V, 6BBS) certified, d in (600, 680)
    (observed 639 A^2), strain norm in (0.09, 0.12) (observed 0.103).
niggli chamber (seeded subsample n = 150; full-set run identical): the
    ball-1 minimiser between each member's Niggli Gram and the medoid's is
    a signed permutation in every case (0 jumps; full set 0/1065); min
    |b - a| over members < 0.5 A (the family edges the a=b exchange wall).
strain mode (same subsample): PCA of the 6-component strain tensors vs the
    medoid has first-mode share > 0.6 (full set 0.79/0.10/0.06); MDS of
    certified ideal distances on a stratified n=112 subsample puts 84% of
    variance on one axis [Figure: fig_ca].
```